State-based test oracles for interrupted agentic workflows

Research-fit statement | Shin Hwei Tan

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International applicant; neither a Canadian citizen nor a permanent resident.

Motivation and fit

Your group's Understanding Bugs in Modern Agentic Frameworks provides a particularly direct bridge from my project experience to an empirical software-engineering study. Its distinction between framework defects and task-level model failures suggests a disciplined way to classify failures before designing tests. I would use Utopia as one inspectable seed system, then test whether selected fault triggers and outcome checks transfer to other frameworks.

Proposed research question

Can independent state-based test oracles and controlled interruption/replay expose false completion and duplicate effects more reliably than agent self-reports or log-only assertions, and which failures transfer across frameworks?

Preparation I can bring

I led architecture, AI-assisted integration and evidence-backed acceptance of Utopia, a completed and frozen multi-device system. Its bounded records cover two Windows execution workers, Android/Web control, fault/replay histories and independently inspectable results. My Android privacy-evidence research provides additional experience in separating observed signals from stronger claims. These artifacts are a starting point for research, not substitutes for a controlled scientific evaluation.

Initial method

Build a small benchmark of benign, verifiable tasks with expected filesystem or application-state outcomes. Categorize failures as framework-level, application-integration or model/task failures before attributing responsibility. Inject interrupted tool calls, changed permissions, stale observations and altered retry order. Compare self-report checks, ordinary logs and independent postcondition oracles. A later phase would implement a narrow detection or repair mechanism only after the empirical study identifies recurring causes. The goal is a research contribution beyond adding more features to my own project.

Evaluation and feasible starting scope

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1. Oracle validity

Compare completion reports with external state checks using independently labeled cases. Measure false-success detection, missed failures, agreement and the cost of building each oracle; audit oracle bugs separately.

2. Fault-trigger transfer

Port a selected set of interruption, configuration and ordering triggers to two or more frameworks. Measure reproducibility, failure yield and transfer rate, keeping framework versions and tasks fixed.

3. Mechanism ablations

Remove state provenance, weaken checkpoints or alter retry ordering in a sandbox. Measure duplicate effects, permission violations, regression-detection rate and runtime overhead; retain negative and inconclusive results.

First 12 weeks after project agreement

Weeks 1-3: reproduce the relevant taxonomy on a small audited sample and define outcome oracles. Weeks 4-7: construct the controlled fault/replay corpus and baseline checks. Weeks 8-12: run a cross-framework pilot, assess annotation agreement and decide whether a targeted repair mechanism is warranted.

Scope and evidence boundaries

Utopia is a completed engineering baseline, not proof of general agent reliability or a peer-reviewed result. No unrestricted natural-language autonomy, production high availability or deployed wearable assistant is claimed. Public release would use benign tasks and remove credentials, personal traces and nonredistributable data.

Selected sources and artifacts

Official CS PhD supervision listing

Current recruitment and research

Agentic-framework bug study

Utopia implementation and bounded acceptance evidence

Prepared 25 October 2026. Proposed experiments are not completed results. Topic and supervision arrangements remain open for discussion.